

Index audit of the strict-real-time R-peak predictor: paper Eqs. (10)–(11) versus the Python implementation

Working note for the CFA ear-EEG paper

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Scope. This note fixes the meaning of every index in Eqs. (10)–(11) of `paper.tex` as they currently stand, then reads the three Python code paths that implement them and checks them term by term. Conclusion up front: the convention implied by Eqs. (10b), (11c) and (11d) is self-consistent and is exactly the one the code uses; the one place where paper and code part company is the *residual driving the proportional (phase) path*, Eq. (11c), which the code forms against the published anchor while the paper forms it against the previous detected peak. For $K_p = 1$ they coincide; for $K_p < 1$ they do not, and the paper’s version leaves the anchor offset unclosed. Section 4 derives this and gives the one-symbol repair.

1 The equations as they currently stand

With $z_k = t_k - t_{k-1}$ the interval between *arrived* R-peaks,

$$T_{k+1} = T_k + w_{k+1}, \quad w_{k+1} \sim (0, q), \quad (1a)$$

$$z_k = T_k + v_k, \quad v_k \sim (0, r), \quad (1b)$$

and the PI tracker

$$\frac{d\hat{\phi}}{dt} = \frac{1}{\hat{T}_k}, \quad (2a)$$

$$\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}, \quad (2b)$$

$$e_{\phi,k} = \frac{z_k - \hat{T}_k}{\hat{T}_k}, \quad (2c)$$

$$\hat{T}_{k+1} \leftarrow \hat{T}_k + K_{ss}(z_k - \hat{T}_k). \quad (2d)$$

1.1 What the indices must therefore mean

Eqs. (2c)–(2d) compare the measurement z_k against $\hat{T}_k$ and emit $\hat{T}_{k+1}$. That forces one reading, and only one:

$\hat{T}_k$ is the one-step-ahead *prior* estimate of the k -th interval, formed from data through beat $k - 1$.

(3)

Two consequences worth stating explicitly, because they are what make the block consistent.

1. **(2d) is a combined update-and-predict step, not a plain Kalman update.** Under the random walk (1a) the prior for T_{k+1} equals the posterior for T_k , so “correct $\hat{T}_k$ with z_k ” and “propagate to beat $k+1$ ” are the same arithmetic. Writing the left-hand side as $\hat{T}_{k+1}$ is therefore correct and is what the code does in a single line.

Table 1: Reading of each symbol under convention (3). “Available at” means the earliest instant a causal implementation can evaluate it.

Symbol	Meaning	Available at
T_k	true k -th RR interval, $t_k - t_{k-1}$	never (latent)
z_k	measured k -th interval from arrived peaks	t_k
$\hat{T}_k$	prior estimate of T_k , from data through beat $k - 1$	t_{k-1}
$\hat{t}_k$	predicted k -th peak, $\hat{t}_{k-1} + \hat{T}_k$	t_{k-1}
$e_{\phi,k}$	normalized phase error observed at the k -th peak	t_k
$\hat{T}_{k+1}$	prior estimate of T_{k+1} = posterior of T_k	t_k
P_k^-	prior variance of $\hat{T}_k$	t_{k-1}

2. **(2a) is the coast rate over the interval that *ends* at t_k** , i.e. valid for $t \in [t_{k-1}, t_k)$: it is the reciprocal of the prior for the interval currently in progress. Instantiated one beat later, the rate in force during the strict-real-time window before t_{k+1} is $1/\hat{T}_{k+1}$ — which is precisely the quantity (2d) produces at the arrival of t_k . So the four lines chain correctly as a generic per-beat recursion.

Chronological order of execution at beat k is (2a) over the whole interval, then at the arrival of t_k : (2c), (2b), (2d). The subequation order in the paper is a, b, c, d ; only b and c are transposed relative to execution, which is harmless since c is a definition.

2 The three code paths

2.1 kalman_predict_rpeaks — the scalar Kalman predictor

src/ekg_iloc_rt.py, lines 281–290:

```

281 for k in range(1, observed.size):
282     predicted[k] = observed[k - 1] + rr_hat
283     measured_rr = observed[k] - observed[k - 1]
284     prior_covariance = covariance + q_samples
285     sigma_tau[k] = np.sqrt(prior_covariance)
286     innovations[k] = (measured_rr - rr_hat) / np.sqrt(prior_covariance + r_samples)
287     gain = prior_covariance / (prior_covariance + r_samples)
288     rr_hat = rr_hat + gain * (measured_rr - rr_hat)
289     covariance = (1.0 - gain) * prior_covariance

```

The loop-carried variable `rr_hat` is read *before* it is written, so at the top of iteration k it holds $\hat{T}_k$ and at the bottom it holds $\hat{T}_{k+1}$ — convention (3) verbatim. Initialization `rr_hat = observed[1]-observed[0] =  $z_1$` seeds $\hat{T}_1$, and `predicted[1] = observed[0] + rr_hat` is $\hat{t}_1 = t_0 + \hat{T}_1$. Likewise `covariance` holds the posterior and `prior_covariance =  $P_k^-$` .

This path has *no* explicit K_p : the anchor is rebuilt on `observed[k-1]`, the arrived peak, every beat. That is a hard phase reset, i.e. $K_p = 1$ in (2b).

2.2 run_as_built — the shipped PI structure

src/pll_design.py, lines 63–69:

```

63 for k in range(1, len(t)):
64     pred = that + That
65     e[k - 1] = pred - t[k]

```

```

66     meas = t[k] + v[k]
67     that = pred + kp * (meas - pred)          # only kp of the error removed
68     if k >= 2:
69         That = That + alpha * ((meas - (t[k - 1] + v[k - 1])) - That)

```

Here $\text{That} = \hat{T}_k$ on entry, that is the *published* anchor $\hat{t}_{k-1}$ after correction, and $\text{pred} = \hat{t}_k^- = \hat{t}_{k-1} + \hat{T}_k$. The integral path (line 69) updates from $\text{meas} - \text{prev_meas}$, which is exactly the paper's $z_k = t_k - t_{k-1}$ (jitter included), with $K_{ss} = \text{alpha}$. The proportional path (line 67) uses $\text{meas} - \text{pred}$, the *anchor* residual. These are two different residuals; Section 4 is about that.

2.3 causal_modes.py — the strict arm that actually cancels

src/causal_modes.py, lines 82–125 (abridged):

```

for k in range(len(r_samples)):
    ...
    t_pred = r_samples[k - 1] / fs + T_hat          # line 94
    w_pred = (T_hat / T_bar) ** gamma              # line 96
    ...
    preds["strict"] = np.interp((ts - fid_pred / fs) / w_pred, c_grid, m["strict"])
    ...
    T_hat += ewma * (Tk - T_hat)                    # line 125, end of iteration

```

Same convention again, and this is the one that matters for Eq. (9): both the *anchor* (t_pred , line 94) and the *phase rate* (w_pred , line 96 — the period that de-normalizes $\hat{\phi}$) are taken from the T_hat that is the prior for the beat being predicted, because the update on line 125 is deferred to the end of the iteration. This confirms item 2 of Sec. 1.1: the rate in force while predicting beat k is $1/\hat{T}_k$, hence $1/\hat{T}_{k+1}$ while predicting beat $k + 1$.

3 Term-by-term correspondence

Table 2: Eqs. (1)–(2) against the implementation. ✓ = exact match.

Paper	Code	File:line	
(1a) $T_{k+1} = T_k + w_{k+1}$	prior_covariance = covariance + q_samples	ekg_ilc_rt.py:284	✓
(1b) $z_k = T_k + v_k$	measured_rr = observed[k] - observed[k-1]	ekg_ilc_rt.py:283	✓
$\hat{t}_k = \hat{t}_{k-1} + \hat{T}_k$	predicted[k] = observed[k-1] + rr_hat	ekg_ilc_rt.py:282	✓
(2a) $d\hat{\phi}/dt = 1/\hat{T}_k$	w_pred = (T_hat / T_bar) ** gamma	causal_modes.py:96	✓
(2b) $\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}$	that = pred + kp * (meas - pred)	pll_design.py:67	see §4
(2c) $e_{\phi,k} = (z_k - \hat{T}_k)/\hat{T}_k$	(meas - prev_meas) - That (I-path only)	pll_design.py:69	see §4
(2d) $\hat{T}_{k+1} = \hat{T}_k + K_{ss}(z_k - \hat{T}_k)$	rr_hat = rr_hat + gain * (measured_rr - rr_hat)	ekg_ilc_rt.py:288	✓
K_{ss}	gain = prior_cov / (prior_cov + r_samples); alpha	ekg_ilc_rt.py:287	✓
K_p	kp (= 0.30); implicit 1 in the Kalman path	pll_design.py:67	✓
(12) $\nu = (z - \hat{T})/\sqrt{P^- + r}$	(measured_rr - rr_hat)/sqrt(prior_cov + r_samples)	ekg_ilc_rt.py:286	index, §5

Sign check on (2b): a late beat gives $z_k > \hat{T}_k$, so $e_{\phi,k} > 0$; the running phase at the true peak has

already passed 1, and $\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}$ pulls it back. Correct as written, and $K_p = 1$ reproduces the hard reset of `kalman_predict_rpeaks`.

4 The one real discrepancy: which residual drives the P path

Let $a_k \triangleq \hat{t}_k - t_k$ be the leftover offset of the *published* anchor after the beat- k correction, and let $\hat{t}_k^- = \hat{t}_{k-1} + \hat{T}_k$ be the pre-correction prediction. The code's residual on line 67 is

$$\rho_k \triangleq t_k - \hat{t}_k^- = (t_k - t_{k-1}) - a_{k-1} - \hat{T}_k = (z_k - \hat{T}_k) - a_{k-1}, \quad (4)$$

whereas the paper's (2c) uses $\hat{T}_k e_{\phi,k} = z_k - \hat{T}_k$, i.e. (4) *without* the a_{k-1} term. (Phase converts to time through the anchor: with $\hat{\phi} = (t - A)/\hat{T}_k$, a correction $\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}$ is a shift $A \leftarrow A + K_p \hat{T}_k e_{\phi,k}$.) Propagating each:

$$\text{code: } \hat{t}_k = \hat{t}_k^- + K_p \rho_k \implies a_k = -(1 - K_p) \rho_k = -(1 - K_p) [(z_k - \hat{T}_k) - a_{k-1}], \quad (5)$$

$$\text{paper: } \hat{t}_k = \hat{t}_k^- + K_p (z_k - \hat{T}_k) \implies a_k = a_{k-1} - (1 - K_p) (z_k - \hat{T}_k). \quad (6)$$

Eq. (5) is a first-order loop in a_k with pole $(1 - K_p)$: stable for $0 < K_p < 2$, and $a_k \rightarrow 0$ once the innovations whiten. Eq. (6) has pole 1: the anchor offset *integrates* the innovation sequence and random-walks away. The two agree only at $K_p = 1$, where both give $a_k \equiv 0$.

So (2c) as literally written does not close the phase loop unless $K_p = 1$, and the shipped gain is $K_p = 0.30$. The cause is that one symbol z_k is being asked to serve both paths, but the code deliberately uses different references for them: the I path measures *peak to peak* (so its innovation is the RR innovation the Kalman gain was designed for), while the P path measures *prediction to peak* (so it sees, and removes, accumulated anchor error).

4.1 Minimal repair

Define the P-path error from the running phase at the instant the peak arrives rather than from z_k alone. At $t = t_k^-$ the tracker's phase is $\hat{\phi}(t_k^-) = (t_k - \hat{t}_{k-1})/\hat{T}_k$, so

$$e_{\phi,k} = \hat{\phi}(t_k^-) - 1 = \frac{t_k - \hat{t}_{k-1}}{\hat{T}_k} - 1 = \frac{t_k - \hat{t}_k^-}{\hat{T}_k}, \quad (7)$$

which is $\rho_k/\hat{T}_k$ and matches line 67 exactly. Note (7) reduces to the current (2c) whenever the previous anchor was exact ($\hat{t}_{k-1} = t_{k-1}$), so nothing else in the section changes, and the I path keeps $z_k - \hat{T}_k$ as in (2d). One display line, no renumbering.

5 Remaining index/notation items

1. **Eq. (12) is correct but one beat later than (2c).** Under (3), $\hat{T}_{k+1}$ is the prior for T_{k+1} , so $\nu_{k+1} = (z_{k+1} - \hat{T}_{k+1})/\sqrt{P_{k+1}^- + r}$ is a valid innovation. But its numerator is the same quantity as (2c)'s, evaluated at $k + 1$, and the code computes it at k (line 286) alongside the gain that (2d) uses. Writing $\nu_k = (z_k - \hat{T}_k)/\sqrt{P_k^- + r}$ makes the shared innovation visible and puts the gate at the same instant as the phase correction.

2. **$\hat{T}_k$ is defined twice.** The line before Eq. (5) sets $\hat{T}_k = \hat{t}_k - t_{k-1}$, which is the *measured* interval, i.e. z_k — not a prior estimate. Eq. (5)’s pre-arrival branch needs the predictor output. (Also a missing hat on t_{k-1} there.)
3. **Prediction window notation.** The text writes $t \in [t_{k+1} - cT_k, t_{k+1})$; the note above Eq. (10) already writes $[\hat{t}_{k+1} - c\hat{T}_{k+1}, \hat{t}_{k+1})$, which is the causal form and carries the right index on the period.
4. **Gain symbol case.** $\$k.p\$$ appears twice in the note after Eq. (11) where (2b) defines K_p .
5. **The $K_p=K_{ss}=0$ fallback claim.** With $K_p = 0$ the anchor is never re-referenced to a detected peak — (5) gives $a_k = a_{k-1} - (z_k - \hat{T}_k)$, a free-running oscillator. “Re-use previous R-peak anchored phase” is $K_p = 1, K_{ss} = 0$: hard re-anchor each beat, frozen period. That is also the regime `kalman_predict_rpeaks` implements.